TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group IIIA (Al, Fe, Cr)

The ammonia-group hydroxides — separating aluminium, iron and chromium by their behaviour toward sodium hydroxide and oxidation, with each one's confirmatory tests.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 7 · LEVEL 1 TERM 1

1. Aim and the members' signatures

Aim: to analyse the Group-IIIA precipitate — the hydroxides thrown down by NH_4OH in presence of NH_4Cl (Experiment 4) — and identify Fe^{3+}, Al^{3+} and Cr^{3+}. Colour carries the first reading again:

HYDROXIDE	APPEARANCE
$Fe(OH)_3$	Reddish-brown, gelatinous
$Al(OH)_3$	White, gelatinous
$Cr(OH)_3$	Green (grey-green), gelatinous

A brown precipitate announces iron; white or green precipitates need the separation below, since mixtures blend (brown dominates and can hide the others entirely — hence dissolution and separation are done even when iron is obvious).

2. The separation

1. Dissolve the washed group precipitate in dilute HCl. Add NaOH solution **in excess** and a little bromine water or hydrogen peroxide, and **boil**:
   — $Fe(OH)_3$ precipitates again and stays (iron is not amphoteric).
   — $Al(OH)_3$ dissolves as aluminate and stays dissolved.
   — Chromium is oxidised to yellow **chromate** (CrO_4^{2-}) and stays in solution — the yellow colour of the filtrate is itself the chromium announcement.
2. Filter. **Residue:** $Fe(OH)_3$ — dissolve in dilute HCl and confirm iron (§3).
3. **Filtrate** (aluminate + chromate): divide.
   (a) For aluminium: acidify a portion with dilute HCl, then add NH_4OH (or boil with NH_4Cl): white gelatinous $Al(OH)_3$ reprecipitates — confirm (§3).
   (b) For chromium: the yellow colour; acidify another portion with acetic acid and add lead acetate: yellow precipitate ($PbCrO_4$) confirms chromate, hence chromium.

3. Confirmatory tests

CATION	CONFIRMATIONS
Fe^{3+}	Potassium thiocyanate (KCNS) → blood-red colour. Potassium ferrocyanide → deep blue (Prussian blue) precipitate. (If iron were Fe^{2+}: ferricyanide gives the deep blue and thiocyanate gives no red until oxidised — the pair of tests also names the oxidation state.)
Al^{3+}	White gelatinous hydroxide dissolving in NaOH and reprecipitated by NH_4Cl/NH_4OH from the acidified aluminate solution. Aluminon reagent gives a bright red lake with the hydroxide — the elegant confirmation where available.
Cr^{3+}	Oxidation to yellow chromate (above), then: acetic acid + lead acetate → yellow $PbCrO_4$; or acidify and add H_2O_2 → a transient deep blue colour (perchromic acid) passing into the ether layer if ether is present.

4. Notes

- The boiling with excess NaOH must be thorough: un-oxidised chromium stays as green hydroxide mixed with the iron residue and is reported as absent — the classic Group-IIIA false negative.
- A purely white group precipitate already excludes iron and chromium in quantity; a brown one proves nothing about the others until separated.
- Gelatinous precipitates filter slowly: use fluted paper, wash with hot water containing a little NH_4Cl, and be patient — forcing them through by pressure ruins the filtrate chain.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A Group-IIIA precipitate is white and gelatinous. Which cation is indicated, and which two are effectively excluded?
2. In the NaOH/peroxide separation, where does each of Fe, Al and Cr end up, and in what form?
3. Why does the filtrate’s yellow colour announce chromium before any test is applied?
4. Give two confirmatory tests for Fe^{3+} and state what each produces.
5. A student skips the boiling in step 1 and reports “chromium absent” from a green-tinged iron residue. Explain the error.
6. How is $Al(OH)_3$ recovered from its aluminate solution, and why does NH_4Cl bring it down?

Answer key

1. Al^{3+} ($Al(OH)_3$). Fe^{3+} (brown) and Cr^{3+} (green) are excluded in any appreciable quantity — their hydroxides' colours would show.
2. Fe: residue, as $Fe(OH)_3$ (not amphoteric, reprecipitated). Al: filtrate, dissolved as aluminate. Cr: filtrate, oxidised to yellow chromate.
3. Because the only yellow species the separation can produce is chromate, CrO_4^{2-}, formed by oxidising Cr(III) in the alkaline boil — the colour is the chromium, chemically transformed but unmistakable.
4. KCNS → blood-red colour (ferric thiocyanate complex). Potassium ferrocyanide → deep blue precipitate (Prussian blue).
5. Without boiling with oxidant, Cr(III) is not converted to soluble chromate; its green hydroxide remains entangled with the iron residue and is masked by the brown. The "absence" is an artefact of the skipped step — chromium must be looked for in the filtrate after proper oxidation.
6. Acidify the aluminate, then add NH_4OH (or boil with NH_4Cl): $Al(OH)_3$ reprecipitates. NH_4Cl works by removing the excess alkali (buffering the solution down) so the amphoteric hydroxide can no longer stay dissolved as aluminate.

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